BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Protocols and Layers

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does a wireless access point do?

2. Why can a mesh survive some broken links?
   A) It has only one path
   B) Every device is a server
   C) It has other paths
   D) Every link is wireless

3. "Fibre carries electrical signals through copper wire." Correct this. Explain why.

4. Desktops and moving tablets need network links. "Wi-Fi needs a cable from each tablet to the switch." Correct this. Which links suit these devices?

Today's objectives

Explain the core ideas of network protocols and layers using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

IP address
Identifies devices/interfaces for routing across IP networks.

MAC address
Identifies a network interface at the data-link level for local communication.

DNS
Resolves a domain name to an IP address.

Protocols and layers
Agreed rules such as HTTP(S), TCP/IP and SMTP support interoperable communication.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

IP address
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Resolves a domain name to an IP address.

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Agreed rules such as HTTP(S), TCP/IP and SMTP support interoperable communication.

I Do - Worked model

A pupil enters a website domain name into a browser. Explain DNS and packet delivery.

- The browser needs the server's IP address.
- DNS provides the mapping from domain name to IP.
- Packets can be routed to the destination across networks.
- The browser and server use suitable protocols, e.g. HTTPS for encrypted web transfer.

We Do - Guided practice

1. Explain how a school laptop sends an email.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the comparison that recognises a secure web protocol.

```
protocol = "HTTPS"
if ... then
print("Secure web")
endif
```

You Do - Independent exam questions

Q2. State the purpose of DNS. [2 marks]

Q3. Distinguish IP and MAC addresses. [4 marks]

Q4. Name a suitable protocol for sending email. [1 mark]

Exit Ticket - True or False?

DNS translates a domain name into an IP address.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It enables wireless devices to connect to the network.

2. C - It has other paths

3. Fibre uses light. It supports fast data transfer over long distances and resists electromagnetic interference. Installation can cost more.

4. Ethernet uses cables. Wi-Fi uses radio, so tablets can move around, but interference can affect it. Choose links to suit the distance and data rate needed.

I Do - Explain the model

The browser needs the server's IP address.

DNS provides the mapping from domain name to IP.

Packets can be routed to the destination across networks.

The browser and server use suitable protocols, e.g. HTTPS for encrypted web transfer.

We Do - Model answers

1. Avoid confusing SMTP with IMAP/POP reception.

Code First - Answer

```
protocol = "HTTPS"
if protocol == "HTTPS" then
print("Secure web")
endif
```

Expected output: Secure web

You Do - Self-mark

Q2: Resolves domain names to IP addresses.

Q3: IP used for logical addressing/routing across networks; MAC identifies a network interface for local data-link forwarding.

Q4: SMTP.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.